

Exploring the Psychometric Properties of the Depression Anxiety Stress Scales - 42 (DASS 42) Across Different Populations Using Publicly Available Data

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Word counter: 2984 (excluding tables, images and references)

Abstract

The Depression Anxiety Stress Scales - 42 (DASS-42) is a widely recognized psychological instrument used to measure the severity of depression, anxiety, and stress. This study investigates the psychometric properties of the DASS-42 using data collected between 2017 and 2019. The analysis focuses on three main aspects: internal consistency, latent factor structure, and mental state prediction based on sociodemographic and personality data. To achieve this, the study follows a systematic approach. Firstly, we evaluate the general factor structure of the DASS-42 in a sample of young adults (ages 20-29) using Confirmatory Factor Analysis (CFA). This step aims to verify the established factor structure and its alignment with the questionnaire's intended measurement model. To further explore the validity of the psychometric factor structure in a different cohort, we conducted an Exploratory Factor Analysis (EFA) on a subsample of older adults (ages 65-90), followed by a Confirmatory Factor Analysis (CFA). Finally, we applied a Random Forest classifier (RF) to examine whether the DASS-42 can effectively identify distinct subgroups of severity related to the mental symptomatology, based on key-features that may play a major role in the development of anxiety-depressive and stress-related symptoms.

1 Introduction

Symptoms concerning the psychological sphere that include depression, anxiety, and stress have been found to be on the rise in last years [1, 2]. The Depression Anxiety Stress Scales (DASS-42) is one of the most widely used psychometric tools for evaluating key negative symptoms related to mood changes [3].

To overcome and implement improvement of the psychological measurement tools typical of this domain, P.F. Lovibond and S.H. Lovibond (1995) developed the DASS-42 to account for Depression, Anxiety, and Stress within a single framework. In their original work, they revealed a 3-factorial structure through the use of EFA and a subsequential CFA.

In this brief report we aim to evaluate and test the psychometric features of DASS-42 in a sample retrieved from a publicly available dataset. Specifically, we aimed to test and compare the psychometric latent structure of the DASS-42 data collected online in a subsample of young adults (20-29) with the current available literature related to the psychometric validation of the DASS-42 in similar subsample. Our second objective was to extend the psychometric properties of the DASS-42 in a subsample of older adults aged between 65 and 90 years. Thirdly, by means of a RF classifier, we intended to explore the discriminative capability of additional key-features (i.e TIPI subscale score, age, education, gender, marital status, family size and average response time per items) in predicting participants mental state.

2 Materials and Methods

2.1 Dataset

Data were retrieved from Kaggle (Depression Anxiety Stress Scales Responses dataset) and Openpsychometrics.org. The dataset includes items, answers, and metadata comprising 39,775 observations over a period spanning from 2017 to 2019. This dataset contains 42 items for the DASS-42, 10 items for the Ten-Item Personality Inventory, a 16-item control checklist, and demographic data about each subject.

2.2 Statistical Analysis

All statistical analysis were performed through Python 3.12.3. Specifically, the EFA was conducted using the Factor Analyzer package [4], while the CFA was conducted through the package semopy [5]. Data cleaning and plotting were conducted using pandas [6, 7], numpy [8], and matplotlib [9] libraries. Firstly, we filter the dataset for retrieving data from a subset of young adults, ranging from 20 to 29 years old. In order to assess and confirm the latent factor structure of the DASS-42, we ran a CFA using a Maximum likelihood (ML) estimation methods. Due to the ordinal nature of the likert scale, we decided to conduct a multivariate normality test (Mardia's test), to check for important deviation from normality. However, for likert scale with multiple observations and large sample size, it is plausible to use a normal approximation [10, 11]

. To assess model fit, we rely on specific indices. Calculated indices include the chi-square (χ^2), Comparative Fit Index (CFI), Root Mean Square Error of Approximation (RMSEA) and the Tucker-Lewis Index (TLI) - [12]. Subsequently, we examined the internal consistency of each subscale by calculating the Cronbach’s alpha [13, 14]. We further decided to include additional measures to assess the construct validity of the DASS-42, such as item-total correlation and domain-total correlation [15]. Finally, we assesses the criterion validity of the DASS-42 by correlating (*Pearson r*) with Emotional Stability and Extraversion subscale of the Ten Item Personality (TIPI) inventory. Specifically, we expected a moderate negative correlation with the Emotional Stability subscale for higher level scored in the DASS-42 and a moderate negative correlation with the Extraversion subscale [16].

Secondly, we decided to explore possible alternatives in the latent factor structure of the DASS-42. To do this, we decided to conduct an EFA on a subgroup of the elderly population (>65). This allows us, through an exploratory approach, to investigate a new possible latent structure as measured by DASS-42. In fact, in the literature, although the three-factor model is the one normally validated and reported, bifactorial or hierarchical models of higher grade have also been tested [3]. EFA was conducted with a ML extraction method, and the numbers of factor to extract was determined by running a parallel analysis [17]. Due to inconsistencies in the results (*see Results section*), we decided to formalize the number of extracted factors by applying the Kaiser–Meyer–Olkin criterion (KMO) (eigenvalue >1). Given our expectation that the factors might not be entirely independent due to the similarity of the psychological constructs measured, we applied oblique rotation (Promax). On the extracted factors, we assessed construct validity through the calculation of the Cronbach’s alpha, the item-total correlation and the domain-total correlation. Moreover, we assessed the criterion validity of the DASS42 correlating each subscales obtained with the Emotional Stability and Extraversion subscale of the TIPI. Afterwards, after the EFA, we ran a CFA to assess the reliability of the results. Importantly, CFA was applied with the same parameters described above.

Thirdly, we aimed to explore the potential predictive role of additional key features reported in the original dataset in mental well-being. Following a similar procedure reported in [18], we decided to apply a Random Forest (RF) classifier to disentangle the relationship between the key characteristics (i.e., TIPI items, age, family size, sexual orientation, marital status, and average response times to the items of the questionnaire) and the mental state of the participants. Specifically, we defined the mental state of the participants based on a cut-off score for each subscale as described in [18]. RF classifier has been applied through scikit-learn library [19].

For the training and test partition, the feature matrix X and the corresponding target vector y were split such that 80% of the data was used for training and 20% was used for testing. An RF classifier was used for this task, with 100 decision trees as a default parameter ($n_estimators = 100$) and a fixed random seed of 42 ($random_state = 42$). The choice of key features as predictors of mental health was made in accordance with studies highlighting the role of sociodemographic and personality related factors in mental health [20–23].

The performance metrics that we considered are accuracy, precision, recall, F1-score, and confusion matrix. Precision measures the ratio of correctly predicted positive instances; recall

99 gives the ratio of the actual positives correctly identified. The F1-score provides the harmonic
100 mean of both precision and recall.

101 **2.2.1 Methodological Clarification: Factor Analysis for Likert type data**

102 It is not the main objective of this assignment to review the current literature on the practical
103 use of likert scales in the field of psychology, so for further discussion please refer to the following
104 papers [10, 24–26]. For practical purposes, and in line with the majority of the literature on the
105 validation of the DASS-42, we chose to align with current practices and treat the Likert scale
106 data as interval [27–31].

107 **2.2.2 Methodological Clarification: Psychometric validation for surveys and ques-** 108 **tionnaires**

109 Additional information about the framework to adopt and the fundamental theoretical concept
110 (e.g construct validity, internal coherence, criterion validity) for validating a psychological test
111 can be found here [32] and here [15].

112 **3 Results**

113 **3.1 The latent structure of the DASS42 on young adults**

114 Mean age of responders was 23.46, while standard deviation (SD) was 2.12. Response mean and
115 SD for each subscale of the DASS42 and for the TIPI are reported in Table 1. The Mardia’s
116 test for multivariate normality was applied, suggesting a significant deviation from multivariate
117 normality ($\chi^2=1.84$, $p < 0.05$). After applying the CFA, the model shows reasonable fit overall.
118 Specifically, CFI index ranges between values of 0 and 1. An index of 0.89, although not excellent,
119 may suggest a reasonable fit. An RMSEA index of 0.05 can be considered as sufficient to show a
120 good fit. Similarly, a TLI of 0.89 could be considered marginal for model fit interpretation (Table
121 2). Taken together, based on the psychometric literature, these indices suggest a sufficient but
122 still marginal fit, leaving room for further discussion of the factorial structure of the DASS42
123 scale.

124 **3.2 Internal consistency and psychometric properties**

125 As described in Table 1, we have assessed internal consistency through the use of Cronbach’s
126 alpha. revealing a consistency of 0.95 for the Depression subscale, 0.90 for the Anxiety subscale,
127 and 0.92 for the Stress subscale. This suggests a very high consistency of the items within each
128 subscale, indicating that all items within each subscale tend to vary together and measure similar
129 facets of the underlying construct. Subsequently, we assessed item-total correlation, reporting
130 levels for all items in each subscale greater than 0.30, which is considered satisfactory [33, 34].
131 We also reported a satisfactory level of domain-total correlation, respectively 0.75 ($p<0.01$) for

the Depression subscale, 0.78 ($p < 0.01$) for the Anxiety subscale, and 0.84 ($p < 0.01$) for the Stress subscale, suggesting a possible interaction between measured constructs (Table 3, left).

Table 1: Means, Standard Deviations, and Alpha Coefficients for the DASS and TIPI subscales

Scale	Subscale	Mean	SD	Cronbach's Alpha
DASS	Depression	2.46	0.86	0.95
	Anxiety	2.11	0.69	0.91
	Stress	2.47	0.73	0.92
TIPI	Emotional Stability	3.31	1.48	0.55
	Extraversion	3.51	1.50	0.46

Table 2: CFA Fit Indices Assessing the Latent Structure of the Data

N	χ^2	DF	CFI	RMSEA	TLI
15792	42562.42, $p < 0.01$	861	0.89	0.05	0.89

Table 3: Domain-Total Correlation

Domain	Domain-Total Correlation	Domain	Domain-Total Correlation
Depression	0.75**	Depression	0.83**
Anxiety	0.78**	Anxiety	0.81**
Stress	0.84**	Stress	0.77**

**Correlation significant at the .01 level (two-tailed).

3.3 The convergent validity of the DASS42 with the TIPI

Further information related to the convergent validity of the DASS42 can be retrieved on Table 4, *Supplementary Figure 1*, and Table 5.

Table 4: Intercorrelations between the DASS42 scales and TIPI subscales

Scale	Depression	Anxiety	Stress	Emotional Stability	Extraversion
DASS Depression	1.000	-	-	-	-
DASS Anxiety	0.671**	1.000	-	-	-
DASS Stress	0.751**	0.804**	1.000	-	-
TIPI Emotional Stability	-0.501**	-0.509**	-0.628**	1.000	-
TIPI Extraversion	-0.277**	-0.165**	-0.173**	0.173**	1.000

* Correlation significant at .05 level (two-tailed); ** Correlation significant at .01 level (two-tailed).

Table 5: Correlations between demographic variables and DASS scores

Demographic variable	Depression	Anxiety	Stress	Total
Education	-0.07**	-0.04**	-0.05**	-0.06**
Gender	0.03**	0.10**	0.11**	0.08**
Age	-0.01**	-0.07**	-0.03**	-0.04**

**Correlation significant at the .01 level (two-tailed).

3.4 Exploring alternatives factor structure

As described in the methods section, we filtered our dataframe for subjects on age between 60 and 90, resulting with a mean age of 66.46 and a standard deviation of 5.33. Then we employed an EFA using ML extraction with an oblique Promax rotation. The number of factors to retain was initially decided by applying a parallel analysis, which yielded unsatisfactory and incongruent results with the current literature regarding the psychometric structure of the DASS-42 in young adults.

Subsequently, to determine the number of factors to retain, we decided to utilize a more well-defined approach based on the DASS-42 literature. Thereby, we employed a scree plot in conjunction with the KMO criterion. The KMO criterion suggested retaining up to three factors (eigenvalue >1), explaining a cumulative variance of 35%. These results appear to partially overlap with the original work of Lovibond and Lovibond (1995), exhibiting the same number of factors and a relatively similar explained variance (in the original work, it reaches 41.3%). However, after examining dissimilarities in the items loading on each factor, we observed some discrepancies. Specifically, the depression subscales fully represented the depression subscale item loadings of the original DASS-42. In this case, we can conclude that the depression subscale was coherent in our analysis. Conversely, for the anxiety and stress subscales, we found only a subset of items loading coherently on each factor as described in the original work loadings. Items exhibiting cross-loadings were identified and compared to the original work [30]. These cross-loading items were subsequently removed from the following analysis due to their inability to account for a significant proportion of variance on a specific factor, which could potentially lead to misleading interpretations of the final DASS-42 scores. After removing the cross-loading items, we conducted a sanity check of each subscale by calculating internal consistency (Cronbach’s Alpha), item-total correlation, and domain-total correlation. The depression subscale demonstrated a Cronbach’s Alpha of 0.96, while the Anxiety and Stress subscales exhibited alphas of 0.88 and 0.92, respectively. Means, Standard Deviations and Alpha coefficients are reported in Table 6.

Subsequently, we evaluated the item-total correlations, finding values greater than 0.30 for all items on each subscale, which is considered satisfactory. Finally, we examine the correlations between the total domain items, reporting values of 0.90 for the depression subscale, 0.86 for the anxiety subscale, and 0.83 for the stress subscale. Data are reported in Table 3, right. On Table

168 8 we reported the correlation between specific demographic variables and the DASS subscale
169 score. The intercorrelation between the DASS-42 items and TIPI can be retrieved from Table
170 7 and *Supplementary - Figure 2*. As a last step, we tested the factorial structure we obtained
171 through CFA, in a similar way to what we did previously. Fit indices reported in Table 9,
172 described a marginal fit. Specifically, we reported a CFI equal to 0.89, a RMSEA equal to 0.07
173 and a TLI of 0.88, pointing toward a suboptimal fit of the factorial structure based on these
174 data.

Table 6: Means, Standard Deviations, and Alpha Coefficients for the DASS and TIPI subscales

Scale	Subscale	Mean	SD	Cronbach's Alpha
DASS	Depression	2.09	0.86	0.96
	Anxiety	1.60	0.62	0.88
	Stress	1.87	0.70	0.92
TIPI	Emotional Stability	4.11	1.85	0.78
	Extraversion	3.37	1.80	0.66

Table 7: Intercorrelations between the DASS42 scales and TIPI subscales

Scale	Depression	Anxiety	Stress	Emotional Stability	Extraversion
DASS Depression	1.000	-	-	-	-
DASS Anxiety	0.64**	1.000	-	-	-
DASS Stress	0.60**	0.65**	1.000	-	-
TIPI Emotional Stability	-0.47**	-0.50**	-0.58**	1.000	-
TIPI Extraversion	-0.34**	-0.17**	-0.13**	0.16**	1.000

* Correlation significant at .05 level (two-tailed); ** Correlation significant at .01 level (two-tailed).

175 3.5 Sociodemographic features as a predictor of Mental State

176 The RF clustering algorithm gave an accuracy of prediction equal to 0.59 for the five mental
177 state category *Normal, Mild, Moderate, Severe, Extremely Severe*. Confusion matrix is reported
178 in Table 10. Although the model showed good predictive ability for an Extremely Severe class
179 mental state (F1-score: 0.75), it struggled with the Moderate (F1-score: 0.45), Mild (F1-score:
180 0.36) and Severe (F1-score: 0.32) categories. Although the total accuracy of the model is superior
181 to random guessing, this suggests to us that there may still be room for improvement through
182 feature engineering, hyperparameter tuning or the choice of an alternative clustering algorithm.
183 Features importance analysis revealed sociodemographic and personality factors may play a role
184 in determining the predictivity of the model. TIPI4, average response time, TIPI9 and familysize
185 appears to be the most relevant in model predictivity.

Table 8: Correlations between demographic variables and DASS scores

Demographic variable	Depression	Anxiety	Stress	Total
Education	-0.09	-0.10	-0.04	-0.11
Gender	0.09	0.06	0.07	0.09
Age	-0.09	-0.13**	-0.14*	-0.13**

* Correlation significant at .05 level (two-tailed); ** Correlation significant at .01 level (two-tailed).

Table 9: CFA Fit Indices Assessing the Latent Structure of the Data

N	χ^2	DF	CFI	RMSEA	TLI
253	42562.42, $p < 0.01$	431	0.89	0.07	0.88

Table 10: Confusion Matrix of the RF model

Actual \ Predict	Extremely Severe	Mild	Moderate	Severe
Extremely Severe	3314	12	242	396
Mild	42	121	235	52
Moderate	401	63	689	362
Severe	1093	16	390	527

4 Discussion

The process of validating psychometric instruments used to measure specific psychological constructs plays a primary role in psychological measurement [32]. These properties should be checked and redefined on different statistical samples, with varying psychometric properties, to ensure greater measurement accuracy and cross-sectionality of the instruments used. In this short assignment, we have approached the validation and exploration of some central properties typically defined in questionnaire validation. Firstly, we aimed to confirm and validate the original factor structure of the DASS-42 through the use of a Confirmatory Factor Analysis (CFA). This was conducted on a large sample of young adults. The results obtained on our sample suggest a sufficient but still marginal fit for the DASS-42 compared to other studies conducted on young adult samples. The main reasons for these discrepancies may be intrinsic to the sample we used. In fact, the publicly available dataset contains responses from subjects with significantly different sociodemographic characteristics, including participants from various nations. However, our results point in the same direction as the current literature on the topic[27–30]

Secondly, our objective was to explore possible alternatives in the latent factor structure through EFA in a sample of elderly. Although we replicated the original trifactorial structure, we found several problems with cross-loadings between items and factors. This problem in the literature has been associated with a similarity between the specific domain measured by the item, which does not allow EFA to associate it with a specific factor [35, 36]. In fact, the

205 complexity of the item is one of the limitations regarding EFA that, if present, can lead to the
206 phenomenon of cross-loadings [33]. In addition, the use of specific rotation methods or factor
207 extractions can lead to an error in estimating the true factorial structure. However, the fit indices
208 obtained from the CFA, along with additional measures of internal consistency, domain total
209 correlations, and intercorrelations between the DASS-42 scales and TIPI subscales, suggest that
210 the instrument demonstrates sufficient psychometric properties. Similar to the findings reported
211 in the young adult sample discussed earlier, the elderly sample also demonstrated an association
212 between depression, anxiety, and stress. In addition, a negative correlation was observed between
213 emotional stability and extraversion, suggesting that these personality traits may play a role in
214 the regulation of psychological distress [16, 37, 38].

215 Finally, RF clustering reveals a superior predictivity to random guessing. However, it seems
216 to be present a room for improvement through various predictive models and the use of additional
217 features.

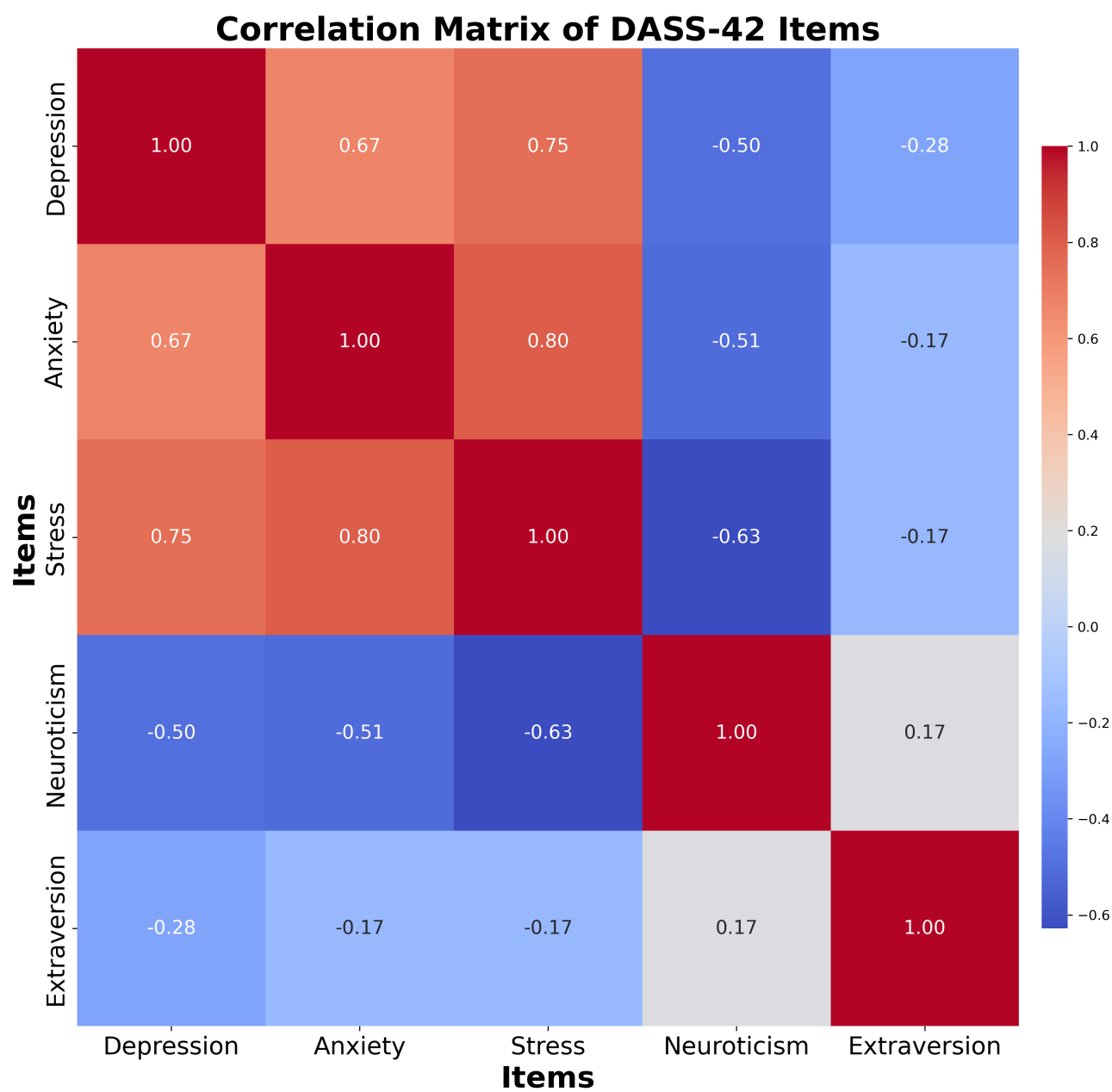


Figure 1: Correlation Matrix

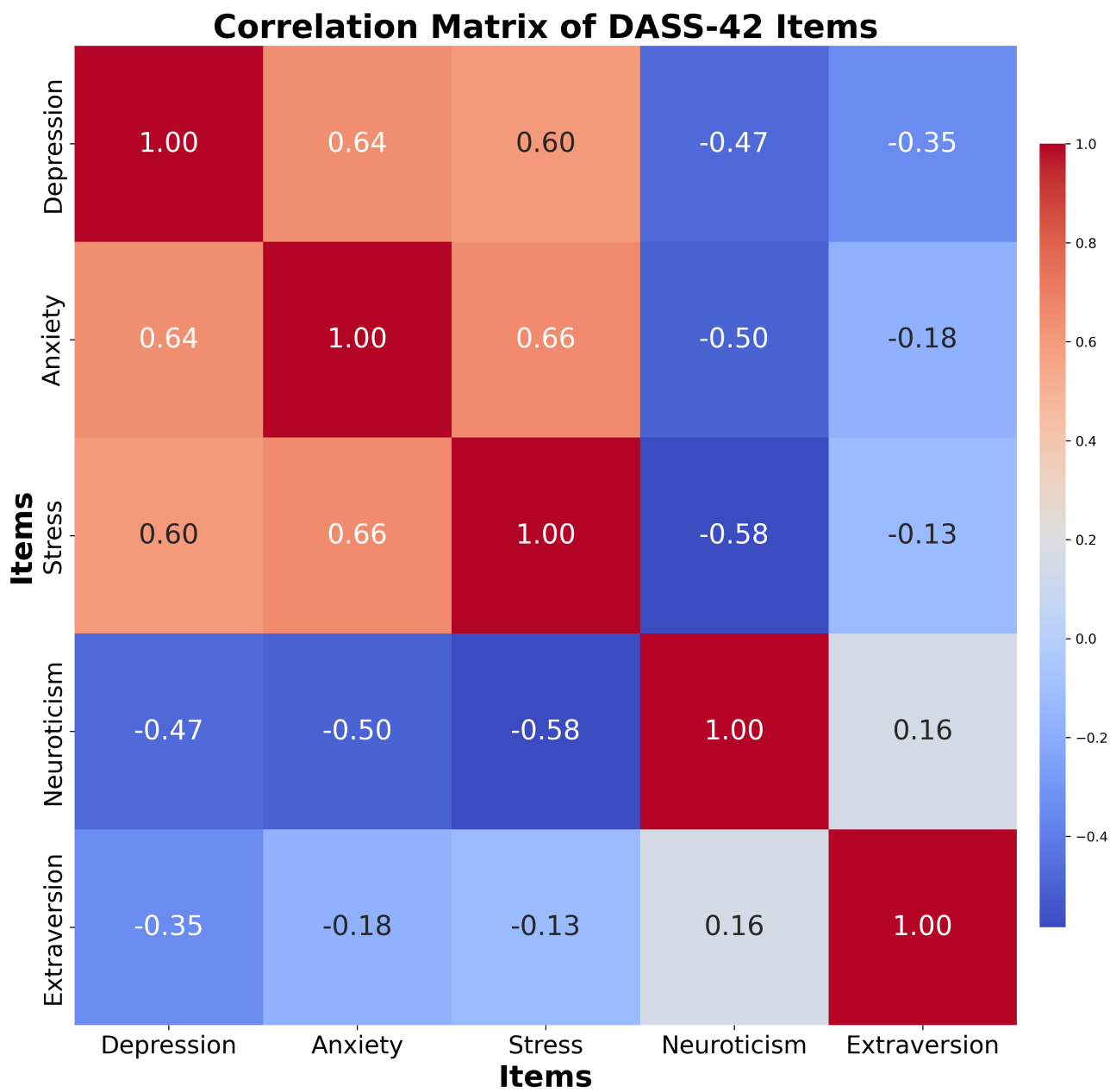


Figure 2: Correlation Matrix

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